

A Technical Introduction to DALL-E image generation

Section 1

Most coverage of DALL-E focuses on a small subset of "surreal" or "quirky" outputs. DALL-E's output for "an illustration of a baby daikon radish in a tutu walking a dog" was mentioned in pieces from Input, NBC, Nature, and other publications. Its output for "an armchair in the shape of an avocado" was also widely covered. ExtremeTech stated "you can ask DALL-E for a picture of a phone or vacuum cleaner from a specified period of time, and it understands how those objects have changed". Engadget also noted its unusual capacity for "understanding how telephones and other objects change over time". According to MIT Technology Review, one of OpenAI's objectives was to "give language models a better grasp of the everyday concepts that humans use to make sense of things". Wall Street investors have had a positive reception of DALL-E 2, with some firms thinking it could represent a turning point for a future multi-trillion dollar industry. By mid-2019, OpenAI had already received over \$1 billion in funding from Microsoft and Khosla Ventures, and in January 2023, following the launch of DALL-E 2 and ChatGPT, received an additional \$10 billion in funding from Microsoft. Japan's anime community has had a negative reaction to DALL-E 2 and similar models. Two arguments are typically presented by artists against the software. The first is that AI art is not art because it is not created by a human with intent. "The juxtaposition of AI-generated images with their own work is degrading and undermines the time and skill that goes into their art. AI-driven image generation tools have been heavily criticized by artists because they are trained on human-made art scraped from the web." The second is the trouble with copyright law and data text-to-image models are trained on. OpenAI has not released information about what dataset(s) were used to train DALL-E 2, inciting concern from some that the work of artists has been used for training without permission. Copyright laws surrounding these topics are inconclusive at the moment. After integrating DALL-E 3 into Bing Chat and ChatGPT, Microsoft and OpenAI faced criticism for excessive content filtering, with critics saying DALL-E had been "lobotomized." The flagging of images generated by prompts such as "man breaks server rack with sledgehammer" was cited as evidence. Over the first days of its launch, filtering was reportedly increased to the point where images generated by some of Bing's own suggested prompts were being blocked. TechRadar argued that leaning too heavily on the side of caution could limit DALL-E's value as a creative tool.

Section 2

Technical limitations DALL-E 2's language understanding has limits. It is sometimes unable to distinguish "A yellow book and a red vase" from "A red book and a yellow vase" or "A panda making latte art" from "Latte art of a panda". It generates images of an astronaut riding a horse when presented with the prompt "a horse riding an astronaut". It also fails to generate the correct images in a variety of circumstances. Requesting more than three objects, negation, numbers, and connected sentences may result in mistakes, and object features may appear on the wrong object. Additional limitations include generating text, ambigrams and other forms of typography, which often results in dream-like gibberish. The model also has a limited capacity to address scientific information, such as astronomy or medical imagery.

Section 3

Technology The first generative pre-trained transformer (GPT) model was initially developed by OpenAI in 2018, using a Transformer architecture. The first iteration, GPT-1, was scaled up to produce GPT-2 in 2019; in 2020, it was scaled up again to produce GPT-3, with 175 billion parameters.

Section 4

Ramesh, Aditya; Pavlov, Mikhail; Goh, Gabriel; Gray, Scott; Voss, Chelsea; Radford, Alec; Chen, Mark; Sutskever, Ilya (26 February 2021). "Zero-Shot Text-to-Image Generation". arXiv:2102.12092 [cs.CV].. The original report on DALL-E. DALL-E 3 System Card DALL-E 3 paper by OpenAI DALL-E 2 website Craiyon website

Section 5

DALL-E 2 DALL-E 2 uses 3.5 billion parameters, a smaller number than its predecessor. Instead of an autoregressive Transformer, DALL-E 2 uses a diffusion model conditioned on CLIP image embeddings, which, during inference, are generated from CLIP text embeddings by a prior model. This is the same architecture as that of Stable Diffusion, released a few months later.

Section 6

History and background DALL-E was revealed by OpenAI in a blog post on 5 January 2021, and uses a version of GPT-3 modified to generate images. On 6 April 2022, OpenAI announced DALL-E 2, a successor designed to generate more realistic images at higher resolutions that "can combine concepts, attributes, and styles". On 20 July 2022, DALL-E 2 entered into a beta phase with invitations sent to 1 million waitlisted individuals; users could generate a certain number of images for free every month and may purchase more. Access had previously been restricted to pre-selected users for a research preview due to concerns about ethics and safety. On 28 September 2022, DALL-E 2 was opened to everyone and the waitlist requirement was

removed. In September 2023, OpenAI announced their latest image model, DALL-E 3, capable of understanding "significantly more nuance and detail" than previous iterations. In early November 2022, OpenAI released DALL-E 2 as an API, allowing developers to integrate the model into their own applications. Microsoft unveiled their implementation of DALL-E 2 in their Designer app and Image Creator tool included in Bing and Microsoft Edge. The API operates on a cost-per-image basis, with prices varying depending on image resolution. Volume discounts are available to companies working with OpenAI's enterprise team. The software's name is a portmanteau of the names of animated robot Pixar character WALL-E and the Spanish surrealist artist Salvador Dalí. In February 2024, OpenAI began adding watermarks to DALL-E generated images, containing metadata in the C2PA (Coalition for Content Provenance and Authenticity) standard promoted by the Content Authenticity Initiative.

Section 7

DALL-E 3 While a technical report was written for DALL-E 3, it does not include training or implementation details of the model, instead focusing on the improved prompt following capabilities developed for DALL-E 3.

Section 8

Given an existing image, DALL-E 2 and DALL-E 3 can produce "variations" of the image as individual outputs based on the original, as well as edit the image to modify or expand upon it. The "inpainting" and "outpainting" abilities of these models use context from an image to fill in missing areas using a medium consistent with the original, following a given prompt. For example, this can be used to insert a new subject into an image, or expand an image beyond its original borders. According to OpenAI, "Outpainting takes into account the image's existing visual elements — including shadows, reflections, and textures — to maintain the context of the original image."